BACKGROUND PAPER

Land for Solar and Wind Development on Oʻahu: Use Rules, Ownership, Grid Proximity, and Terrain

Prepared for Michael J. Roberts (UH Mānoa) · July 12, 2026

Compiled by Claude from primary sources — statutes, ordinances, permit dockets, county tax rolls, and GIS layers. Every figure regenerates from cached data (`analysis/`); full citations are in the repository `notes/` files. Items not confirmed against a primary source are tagged U.

Summary. This paper documents the land available for utility-scale solar and wind development on Oʻahu: the use rules that apply to each class of land, who owns it, how far it sits from mapped transmission, and how much of it is flat enough to build on. The main quantitative results: about 40,600 acres of agricultural-district land lie within 1 km of a mapped 46 kV+ line; of that, roughly 6,100–11,100 acres of class D/E land (depending on the slope screen) can host solar today with no acreage cap and no special permit, and a further 3,600 acres of class B/C land are eligible by right under the current 10%-of-parcel/20-acre cap. Where projects exceed the cap, the special-use-permit path has approved 7 of 8 applications since 2014, unanimously, with no intervenors, in a median of about six months. Government owns half the agricultural district; Kamehameha Schools owns a quarter of the private half. Wind faces a different constraint: Honolulu's 2025 zoning ordinance sets large-turbine setbacks of at least 1.25 miles from residential-district lots, which leaves 36% of AG/Country-zoned land geometrically eligible and bars repowering of the existing fleet.

1 Use rules by land class

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Hawaiʻi's Land Use Law places every acre in one of four state districts; the agricultural district covers about 276,000 acres of Oʻahu (roughly 47% of the island). Within the district, solar rules key off the Land Study Bureau's soil productivity ratings, A (best) through E. Four layers of law apply to a project: the state district rules (HRS ch. 205), county zoning (the Honolulu Land Use Ordinance), county real-property taxation, and — for any project selling to the grid — a power-purchase agreement with Hawaiian Electric approved by the PUC. Table 1 states the current state-district rules for solar; wind is covered in §8.

Table 1. **Current state-district rules for solar on agricultural land (HRS §§205 (20)–(21))**

Soil class	Solar as a permitted use	Above the cap
A	Not permitted	No SUP path: the §205-4.5(a)(20) proviso bars sola class A outright, and (a)(21) authorizes SUPs on B/ No solar SUP on A-rated land has ever been grante applicants carve A soils out of project areas. Reclassification requires an LUC district boundary amendment
B / C	Up to min(10% of parcel, 20 acres) per parcel	County special use permit (§205-6), with condition paneled area must be offered for compatible agricul use at ≥50% below fair-market rent; decommissioni security; site restoration
D / E	Permitted outright — no acreage cap, no special permit	n/a

Three county-level rules matter alongside the state rules:

- **Zoning.** The Honolulu LUO permits solar in AG districts subject to ordinary development standards; solar faces no county setback regime comparable to wind's (§8).
- **Property tax.** Land under agricultural dedication is assessed at 1–5% of fair-market value. In 2021 the city assessor reclassified land under operating solar farms as industrial (full value, $12.40/$1,000), raising some annual bills about 25-fold mid-contract; an 80% exemption for renewable projects was subsequently enacted. Assessment treatment remains a material, county-discretionary cost item for project underwriting.
- **Procurement.** All grid sales run through HECO power-purchase agreements awarded in competitive RFP rounds and approved by the PUC. In practice land is optioned and permits filed after an award, so procurement timing governs when land actually enters development.

2 Procedural pathways and timelines

A project's permitting path depends on soil class and size. On D/E soils, and on B/C soils within the cap, solar is a permitted use: county building, grading, and electrical permits apply, and no discretionary land-use approval is required. Above the cap on B/C soils, the special-use-permit path runs through the county planning commission and, for permits over 15 acres, the state Land Use Commission. A complete census of the LUC's special-permit series (1963–2026) contains eight solar dockets, all filed since 2014:

**Table 2. Solar special use permits reaching the Land Use Commission, 2014–
census)**

Statistic	Value
Dockets	8 (SP15-405/406/407, SP17-408, SP21-411/412, SP26-
Outcomes	7 approved, 1 pending, 0 denied, 0 withdrawn; every LUC
Intervenors	0 of 8; organized testimony in the dockets is supportive (u Ulupono)
Median duration	County stage 141.5 days; LUC stage 34 days; total ≈ 6 mc

Applicants and landowners	AES on 4 of 8; land owned by Kamehameha Schools, Grov McBryde/A&B, Robinson, Mahi Pono, Castle & Cooke, Mon
Counsel	Matsubara Kotake & Tabata on 5 of 8

The SUP conditions are inexpensive in practice. The ag co-use clause requires that the paneled area be *made available* for compatible agriculture at half of market rent; grazing qualifies, and grazing leases substitute for vegetation management the operator would otherwise purchase. Decommissioning security and restoration are standard, modest costs for PV. The observed cost of the SUP path is mainly time (≈6 months of discretionary process) and the uncertainty of a case-by-case approval, which lenders price. To date every applicant has been a major developer on a large landowner's parcel. Smaller landholders on B/C soil have a 20-acre by-right envelope and would need the SUP process to go beyond it; navigating that process is routine for the repeat developers and law firms in the census, and a developer partner would ordinarily carry it for a lessor.

Where the rules bind therefore depends on parcel size and owner capacity together. On parcels under 200 acres the 20-acre term never binds (10% of the parcel is the smaller number); on parcels above 200 acres the 20-acre term binds, but the owners of most such parcels are governments and large private estates for which the SUP process is routine. Neither restriction binds hard unless the SUP is difficult for the particular owner — which points at mid-size private owners of cap-bound parcels, quantified in §3.

3 Ownership

Ownership of the Oʻahu agricultural district is highly concentrated (Figure 1). Attribution covers 99% of the district's acreage, from the city's bulk assessment table with entity resolution across name variants.

- Government owns 50.1%: State of Hawaiʻi 25.8% (including DLNR and ADC), the United States 19.6% (largely military), the city 2.7%, DHHL 2.0%.
- Kamehameha Schools owns 12.8% of the district — 26.1% of the private half. The twenty largest private owners hold 71.2% of private acreage.
- The 20-acre cap term binds only on parcels above 200 acres with B/C soil: 118 parcels on Oʻahu, led by the United States, Kamehameha Schools,

Dole, and the State. Changing the cap therefore changes eligibility mostly on large institutional holdings.

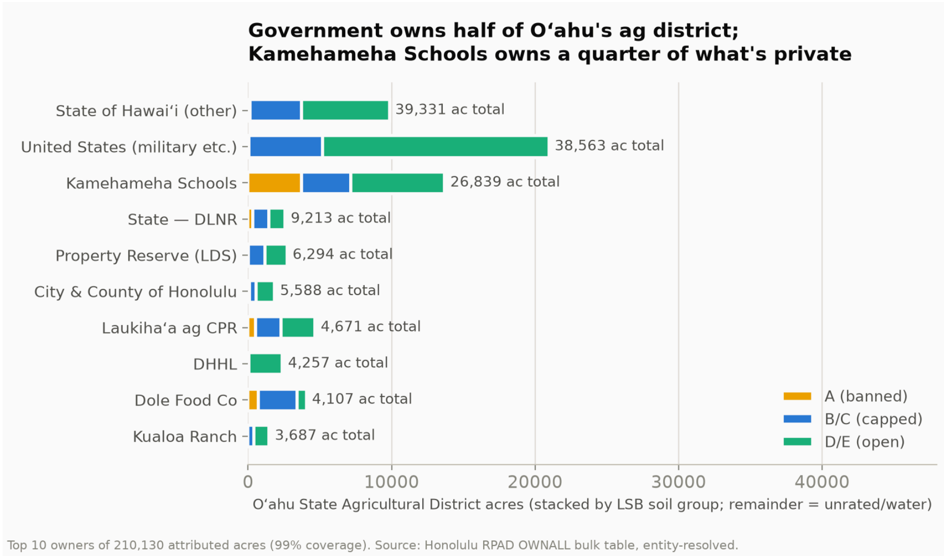

Figure 1. Top-10 owners of the Oʻahu agricultural district, stacked by soil group. Source: Honolulu RPAD OWNALL bulk table joined to parcel–soil intersections; 99% of acreage attributed.

Parcel size × owner scale: where the cap binds and for whom the SUP is hard

Table 3 crosses parcel size against owner scale (an owner's total ag-district holdings, with governments broken out) for B/C acreage. Reading it against the rules in §§1–2:

- **Parcels under 20 acres** (10% term binds; ≤2 buildable acres each): B/C acreage here is mostly small-owner land — suited to accessory or CBRE-scale projects, not utility-scale siting.
- **Parcels of 20–200 acres** (10% term binds; 2–20 buildable acres each): about 6,100 B/C acres across all owner classes. By-right projects up to 20 acres are possible at the top of this range.
- **Parcels over 200 acres** (20-acre term binds): 118 parcels, 25,200 B/C acres. Governments hold 49 of these parcels, large and major private owners (≥1,000 ac total holdings) hold 43 — owners for whom the SUP process is documented routine (§2) or for whom commercial leasing runs through developers in any case.
- **The exposed set: 26 parcels, 26 distinct mid-size private owners** (100–1,000 acres total holdings), about 3,700 B/C acres. These owners face the

20-acre cap and are the least likely to carry a discretionary county-plus-LUC process themselves. No small owner (<100 ac held) owns a cap-bound parcel. In practice a developer partner would ordinarily run the SUP for a lessor — every SUP in the census was developer-driven — so the practical question for this group is access to developer interest, which runs through HECO procurement rounds, rather than the permit process itself.

Table 3. B/C acreage by parcel size and owner scale, Oʻahu ag district (acres)

Parcel size	Government	Major private (>5,000 ac held)	Large private (1,000–5,000)	Mid private (100–1,000)
<20 ac	1,033	115	80	231
20–50 ac	514	141	208	74
50–200 ac	2,117	337	704	1,300
200–1,000 ac	4,835	857	1,679	3,674
>1,000 ac	5,107	3,169	5,876	0

4 Grid proximity

Public line data (HIFLD and OpenStreetMap, cross-validated) maps Oʻahu's 138 kV network well (~325 km, both sources agree) and the 46 kV network poorly (~80 km mapped of several hundred km built), so all distances below are upper bounds — true distances are shorter. Line capacity and substation headroom are confidential and are not screened here; to the extent upgrades can ride existing rights-of-way, distance to an existing corridor is the relevant siting variable.

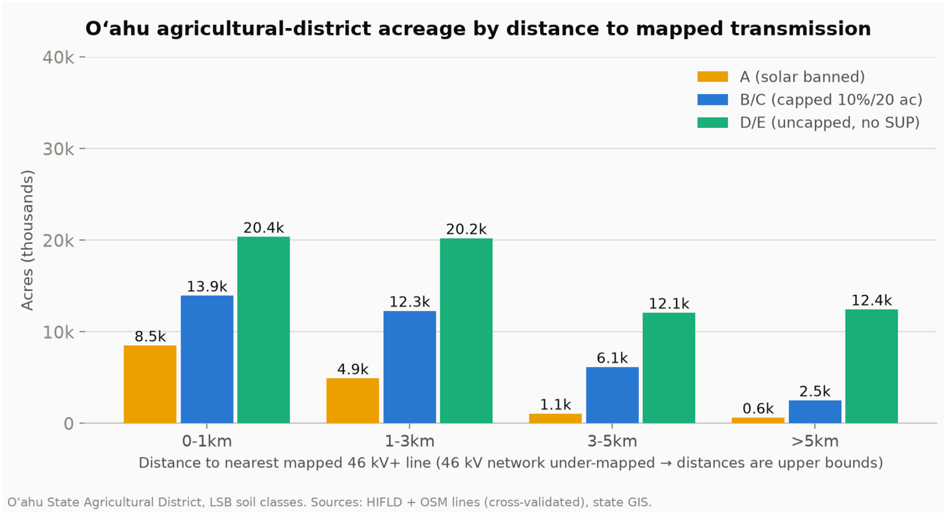

Figure 2. Agricultural-district acreage by distance to the nearest mapped 46 kV+ line and soil group. Within 1 km: 20,359 D/E acres (no cap, no SUP), 13,944 B/C acres (capped), 8,497 class-A acres (solar not permitted). 75% of all B/C acreage lies within 3 km.

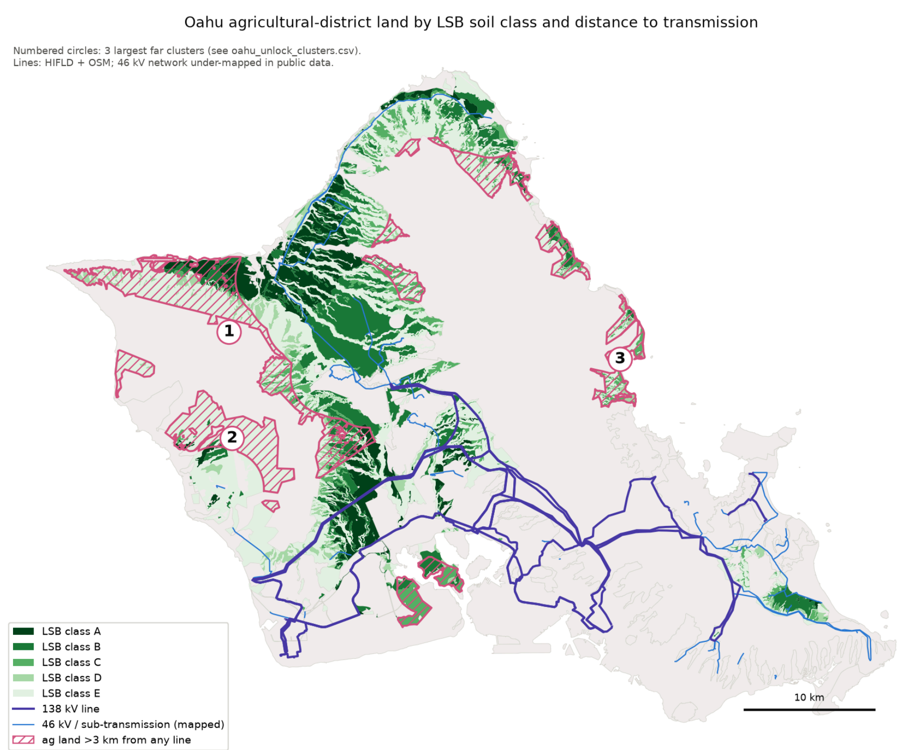

Figure 3. Soil class, mapped lines, and the largest land clusters more than 3 km from any mapped line. The far clusters are Waiʻanae-range hillside (mostly class E), Lualualei valley (largely federal), and Koʻolauloa mauka.

Strategic transmission additions

To locate where new or upgraded lines would add the most access and bidding competition, all buildable (slope-screened) ag-district land more than 1 km from a mapped line was clustered and each cluster scored on two denominators: buildable acres per kilometer of new right-of-way, and distinct landowners served per kilometer (Figure 4, Table 4). Owner count matters for competition: a corridor that reaches five owners' land creates five potential PPA bidders; one that reaches a single estate creates one.

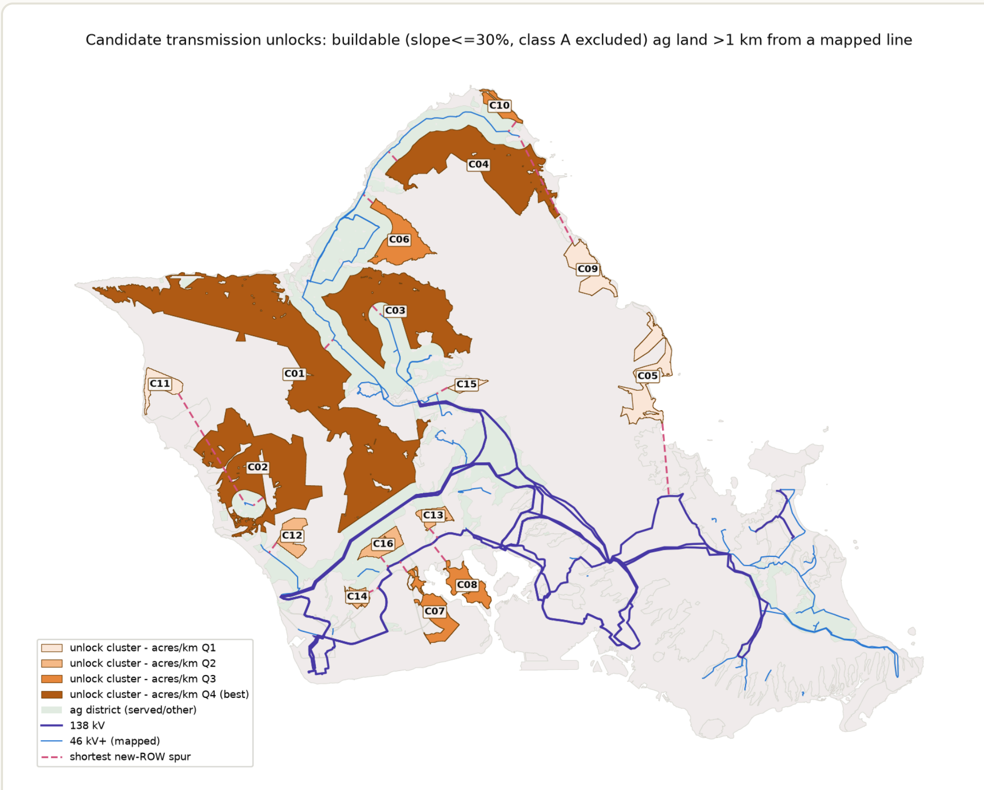

Figure 4. Candidate transmission unlocks: buildable (≤30% slope) ag land beyond 1 km of a mapped line, clustered and shaded by acres-per-km rank, with the shortest new-ROW spur to each cluster (dashed). Class A land is excluded (solar is not permitted on it under any path).

Table 4. Leading corridor candidates and the in-ROW upgrade ring (slope ≤3(MW at 5 ac/MW)

Candidate	Buildable ac	MW	Owners (≥100 ac)	Top owners	Notes
1–3 km ring (upgrade existing ROWs)	≈24,000	≈4,800	28	Dispersed	Served by sh reconductor corridors; la widest owne option. ≤15⁽

C03 Waialua	5,697	1,139	5	Dole, Kamehameha Schools, ag-CPR, State	≈2 km of ne soil, so most contingent The standou competitive
C04 Kahuku	5,630	1,126	4	US 36%, Property Reserve (LDS) 33%	Partly feder politics appl
C01 Waiʻanae mauka/plateau	14,210	2,842	18	Mixed	Largest on military-adja "unserved" artifact of 4 — verify aga before weig
C02 Lualualei	8,749	1,750	—	US Navy (majority)	Majority fed leasable onl enhanced-u

Two readings follow. Upgrading existing rights-of-way serves more buildable land (≈24,000 non-A acres) across more owners (28) than any greenfield corridor, consistent with treating in-ROW capacity upgrades as the first-order transmission strategy. Among genuinely new corridors, a short (~2 km) Waialua spur is the one candidate that combines volume, multiple private owners, and commercial availability — and its value is mostly contingent on relaxing the B/C cap, since 74% of the cluster is B/C soil. Several apparent clusters (Ewa/Puʻuloa, Waiau, Helemano) are likely artifacts of the 46 kV mapping gap, and the federal-majority clusters are available only through military leasing programs. Line and substation capacity, which is confidential, is unexamined throughout.

How much land a given length of new line can reach

A budget-constrained version of the same question: for a total length L of new line, placed optimally (greedy heuristic, branching from the existing mapped network), how much class B–D land at ≤30% slope comes within 1 km of a line? Figure 5 traces the full curve; Figure 6 maps the build-out at the knee. Class E is excluded here along with class A; routes are geometric, so the curve is an upper bound on what any real routing would achieve at a given length, and the under-mapped 46 kV network means baseline coverage is understated.

- Baseline (existing mapped network): 16,328 of 40,870 eligible acres (40%) are within 1 km.
- The marginal payoff is highest in the first ~11 km (≈400–480 acres per km) and declines steadily after; the knee sits near 6 km. Ten kilometers of new line raises coverage to 20,185 acres (+3,857); 25 km to 24,195; 50 km to 29,481; 100 km to 36,103 — 88% of everything eligible.
- The first 6 km are three short spurs, all extensions of existing corridors: ~2.2 km mauka from the Waialua 46 kV loop (the same area as corridor candidate C03), ~2.9 km down the Kunia / central-plateau spine south of Schofield, and ~0.6 km at Kahuku — together +2,252 acres. The 25–50 km range thickens Waialua / Kunia and reaches the Waiawa cluster; spending beyond ~75 km buys Mokuleia and remnants.

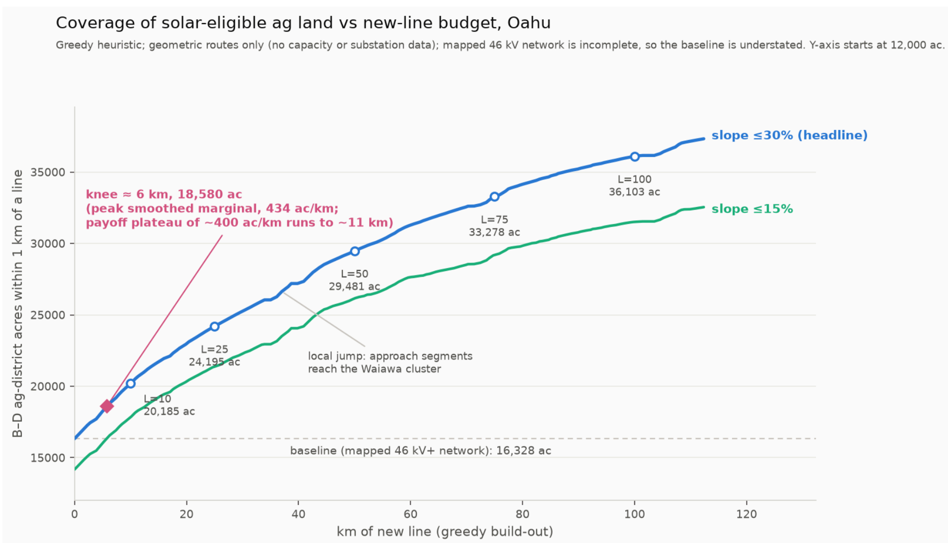

Figure 5. Coverage of B–D ag-district land (≤30% and ≤15% slope) within 1 km of a line, as a function of km of new line placed by the greedy heuristic. Marked points at L = 10/25/50/75/100 km; the knee (≈6 km) is where smoothed marginal gain peaks (434 ac/km).

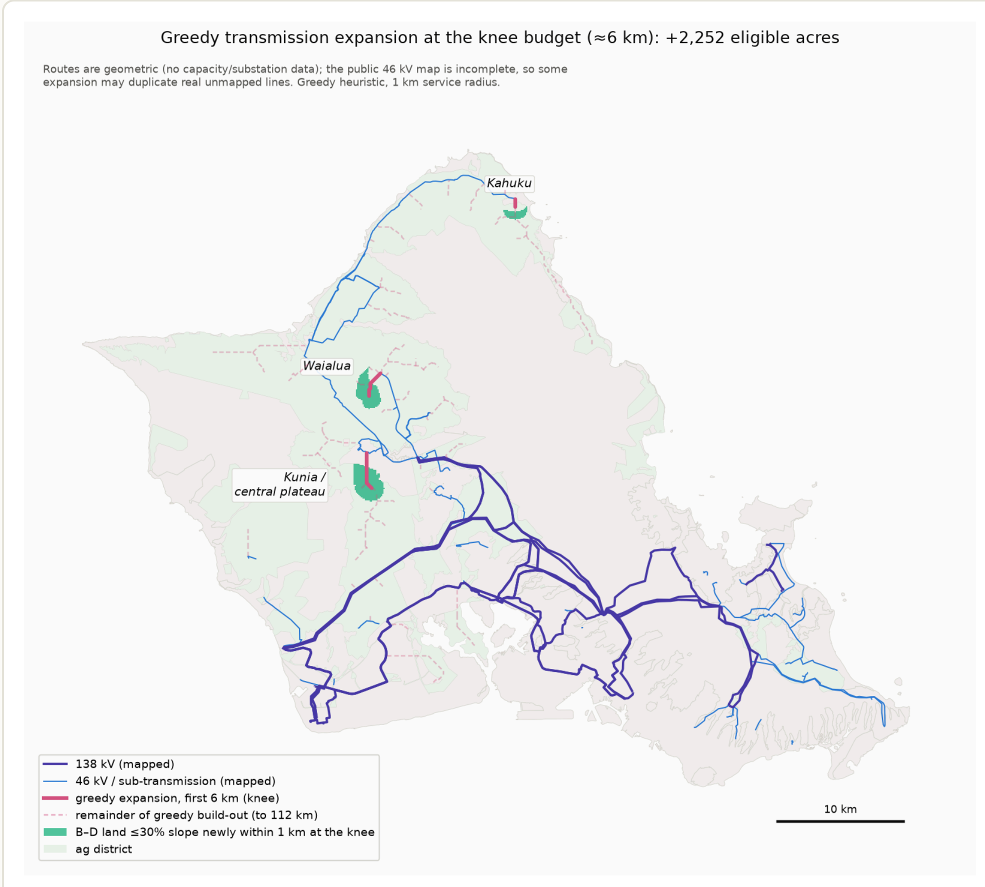

Figure 6. The build-out at the knee budget (≈6 km): three short spurs at Waialua, Kunia, and Kahuku (solid), the remainder of the greedy build-out to 112 km (dashed), and the land newly served (shaded).

Cost benchmarks and an illustrative $500 million program

Planning-level unit costs put the expansion results in dollar terms. Cost per MW-km is not a primitive — it is line cost per km divided by transfer capacity, so it varies an order of magnitude with voltage class. Table 5 gives mainland benchmarks (MISO transmission cost estimation guides, 2024$) with a Hawaiʻi construction multiplier of 1.5–2.5× applied U — HECO's actual unit costs are confidential. Substation work adds fixed costs of several million dollars per connection point, which for spurs as short as the knee build can exceed the wire itself.

Table 5. **Transmission unit-cost benchmarks (mainland planning values × 1.5-multiplier)**

Class	Build cost, Hawaiʻi-adjusted	Typical transfer capacity	Im
46–69 kV sub-transmission	≈ $0.6–2.5M per km	≈ 20–60 MW	≈

138 kV single circuit	≈ $1.5–4M per km	≈ 150–250 MW	≈ $
Substation / collector bay	≈ $10–30M per station	—	fix pc

At these rates, wire length is a second-order project cost: a 3-km spur adds roughly $30–75k/MW to a project whose all-in Hawaiʻi capex is about $2–2.5M/MW — 1.5–3.5%. What a transmission budget mainly buys is *capacity* and *interconnection points*, not route-miles. The greedy curve makes the same point from the land side: about 50 optimally placed km already reaches ~29,500 eligible acres, roughly the ~30,000 acres a fully grid-scale 5–6 GW decarbonization build would need at current densities (later builds at higher MW/acre need less).

Table 6 lays out an illustrative $500 million program built on those observations. Its design principles: rebuild inside existing rights-of-way where possible (new corridors are where organized opposition concentrates); build spurs at 138 kV rather than 46 kV (headroom, not conductor, is the scarce input); and site collector substations by *ownership cluster*, with published available capacity, so that interconnection feasibility becomes visible information and every cluster of independent landowners can produce competing PPA bids.

Table 6. An illustrative $500M transmission program for Oʻahu solar (plannin from Table 5)

Tranche	Budget	What it buys	Rationale from this
1. Rebuild the ag-belt sub-transmission at 138 kV in existing ROWs	≈ $150M	50–70 km: Waialua–Wahiawā–Kunia loop + Kahuku segment	The 1–3 km ring ho buildable non-A acr owners; in-ROW wo permitting exposure 5× the power of 46 more cost
2. Collector substations with pre-built interconnection bays	≈ $120M	5–6 stations: Waialua, Kunia/central plateau, Kahuku, Waiawa/Waipiʻo, ʻEwa edge	Placed by ownershi alone reaches Dole, State land); publish converts private inte information into ope opportunities — the competition in the p
3. The knee spurs, built at 138 kV	≈ $60–80M	20–25 km: Waialua mauka, Kunia spine,	The optimization's k segments (≈400–48

		Kahuku, then toward Waiawa	
4. Backbone delivery reserve	≈ $120M	Second circuit / rebuild on the central 138 kV path, or state cost-share into HECO's Integrated Grid Planning	Collected power mu Honolulu load cente headroom is confide not sized for 5–6 G\ could absorb far mc mandated public ho disclosure at transm
5. Contingency, studies, queue support	≈ $30M	Interconnection studies, hosting-capacity mapping	Study queues are a priced into bids

Two complements the budget cannot substitute for. First, **cap reform**: most of the acreage this network reaches is B/C soil, SUP-only above 20 acres under current law, so the transmission investment and the one-sentence statutory change are strict complements. Second, **storage co-location at the collectors**: grid-forming batteries shift delivery off-peak and shrink the backbone requirement, so part of tranche 4 is plausibly better spent as storage than wire — a dispatch-modeling question beyond this land analysis, and one for a comprehensive capacity-expansion study to settle. On sufficiency: this program removes land access and bid entry as constraints and makes the remaining constraint — bulk delivery — explicit; at these unit costs $500M is transformative for access and competition and only partial for 5–6 GW of deliverability, which is the right order to spend in.

5 Terrain

A 10-meter USGS elevation model, rasterized on the same grid as the parcel, soil, and line layers, adds slope in five-point bands (Figure 7, Table 7). Slope matters differently for the two legal categories. B/C soil is prime agricultural land and is overwhelmingly flat: 89% of the acreage that a relaxed cap would make eligible near the grid sits at or below 15% slope (97% at ≤30%). D/E soil includes large areas of hillside inside the nominal district boundary: 45% of near-grid D/E acreage is steeper than 30%.

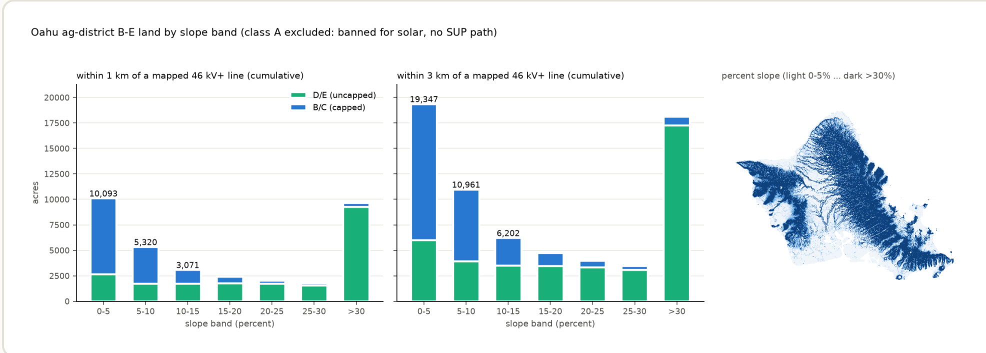

Figure 7. Ag-district land within 1 km and within 3 km of a mapped 46 kV+ line, by percent-slope band and soil group (cumulative panels), with the island-wide slope raster inset. Class A land is excluded.

Table 7. **Buildability and cost by slope band (engineering literature), with Oʻal**

Slope	Standard tracker	Terrain-following tracker	Indicative cost premium vs flat
0–5%	Yes	Yes	≈ 0 (NREL benchmark reference terrain)
5–10%	Yes	Yes	Low single-digit % of capex derived
10–15%	At spec limit (≈15% N–S)	Yes	Mid single-digit % derived; Honolulu grading code requires an engineer's soils report above 15%
15–20%	No	Yes (built precedent ~20%)	Unpublished; erosion/stormwater costs intensify
20–25%	No	Vendor-reported projects planned	Unpublished; anecdotal only
25–30%	No	Within one vendor's 37% spec; no completed project above 25% found	Unpublished; civil, erosion, and O&M dominated
>30%	No	No	Treat as infeasible

The engineering literature contains no published continuous cost-versus-slope curve: NREL's cost benchmarks assume flat sites and its supply curves apply slope as a binary screen (10% reference, 5% restrictive); mainstream terrain-following trackers top out near 15% grade; feasibility claims above

20% rest on a single vendor's specification. A workable modeling convention: ≤15% as the mainstream envelope, 15–30% as a specialized premium-cost margin, >30% infeasible. Aspect effects at 21°N (north- versus south-facing slopes) remain to be quantified with PVWatts/SAM runs.

6 Combined viability estimates

Putting the screens together for land within 1 km of a mapped line (all figures at 5 acres/MW; 7 acres/MW scales them by 5/7):

- **Available today, no discretionary approvals:** D/E soil at ≤15% slope, 6,083 acres (~1,217 MW); at ≤30% slope, 11,110 acres (~2,222 MW). Plus 3,601 acres of by-right B/C eligibility under the current cap (~720 MW, island-wide; spread in ≤20-acre pieces across many parcels).
- **Available via the SUP path:** B/C acreage above the cap, in practice limited by procurement awards rather than by the permit process (Table 2).
- **Under alternative cap rules** (Figure 8): raising the parcel share to 20% while keeping the 20-acre term adds little (4,743 acres); removing the 20-acre term at 10% yields 9,410 acres; 20% with no acreage term yields 15,657 acres (~3,131 MW), of which 72% is within 1 km of a line and ~90% is at ≤15% slope. The 20-acre term, which binds only on parcels over 200 acres, accounts for most of the difference between scenarios.

Combining cap-reformed B/C with buildable D/E gives a land-side envelope for near-grid utility solar on Oʻahu of roughly 16,000–21,000 acres — order 3–4 GW — before flood, aspect, cultural-resource, and interconnection-capacity screens.

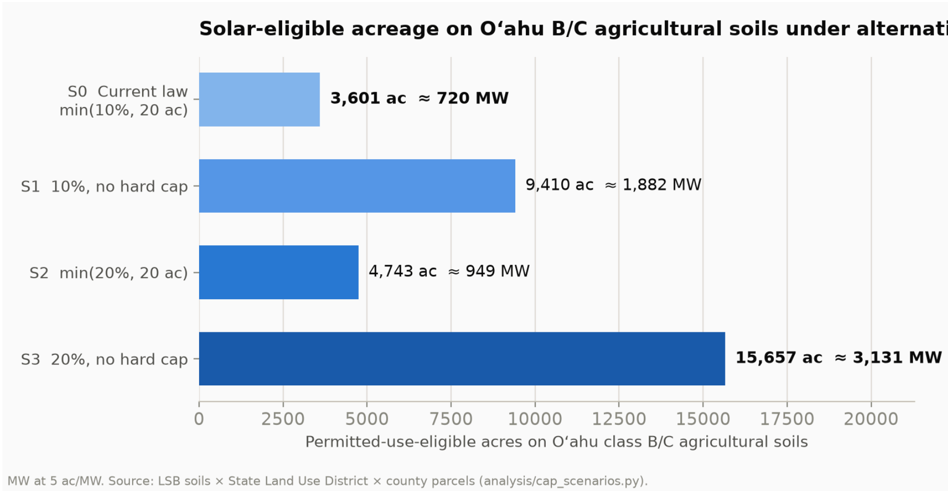

Figure 8. By-right eligible acreage on Oʻahu B/C soils under the current cap and three alternatives. Statewide, the same scenarios run 33,978 → 157,213 acres.

7 Agrivoltaics

Chapter 205 accommodates solar-plus-agriculture at every tier. Within the by-right cap, solar and farming can share a parcel with no further approvals. On the SUP path, ag co-use of the paneled area is a permit *condition* — the statute requires the area be offered for compatible agricultural activities at half of market rent, and grazing, cropping, and apiary uses all qualify. On D/E soils there is no co-use requirement at all. Working examples exist (HARC's 230-acre Mililani agrisolar site has cultivated crops under panels since 2022), and no statutory provision distinguishes agrivoltaic from conventional PV design. The practical constraints on agrivoltaics are economic — racking height, row spacing, and operations coordination — plus the county assessment question in §1: whether co-farmed solar land retains agricultural dedication for the non-paneled area is an administrative practice worth confirming parcel by parcel.

8 Wind siting

The state-district rules for wind are permissive: wind energy facilities have been a permitted use on agricultural land of every soil class since 1980, with no acreage cap, subject to a compatibility proviso. The operative constraint is county zoning. Honolulu's Ordinance 25-2 (effective January 2025) sets these standards for turbines of 100 kW and larger:

- Setback of 1× tip height from all property lines, plus the greater of 10× tip height or 1.25 miles from any lot in a country, residential, apartment, apartment mixed-use, or resort district;
- No director's waivers for wind facilities;
- Repowering limited to a 7% height increase (575 ft maximum), and only within the term of the facility's current power-purchase agreement;
- Shadow-flicker (≤10 hr/yr) and noise (≤10 dBA over ambient) limits at pre-existing occupied buildings.

Geometrically, the 1.25-mile rule leaves 36% of Oʻahu's AG/Country-zoned land eligible, versus 85% under the prior 1×-tip-height standard. Eighteen of Kahuku's twenty turbines are nonconforming under the new rule; because nonconforming turbines cannot repower beyond their current PPAs, the island's ~155 MW of existing wind is expected to retire as contracts end between 2031 and 2040. For land-screening purposes, large-scale wind on Oʻahu should be treated as foreclosed under current county law; the eligible remainder is concentrated in mauka areas away from residential-district boundaries, where slope and access impose their own limits.

9 State and federal land

Because government owns half the district, public-land disposition is a large share of the siting question. The state's leasing authority is broad: HRS §171-95 (2002) authorizes non-auction leases of up to 65 years to renewable energy producers; §171-95.3 (2009) adds a direct-negotiation process; Act 53 (2024) removed the requirement that output be sold to a utility; and ADC is exempt from public-auction requirements. Use to date, Oʻahu:

- DHHL: the most active state-side lessor (Kalaeloa Solar Two, 2013; subsequent leases at Barbers Point and on Kauaʻi).
- University of Hawaiʻi: 12.5 MW at West Oʻahu (AES, 2024).
- DLNR: its first Oʻahu solar disposition under §171-95.3 (Eurus, 20 MW at Olai) reached Board of Land and Natural Resources hearings in January 2026.
- ADC: hosts solar on Kauaʻi; its Oʻahu holdings (1,700+ idle acres per State Audit 21-01, in the Wahiawā/ Whitmore area) have no solar dispositions on record.
- Federal: the military hosts projects under enhanced-use leases (Kupono Solar, ~42 MW on Navy land at Pearl Harbor; HECO's Schofield plant on

Army land). Federal parcels also hold the largest block of flat, near-grid D/E land in the screen (Lualualei).

Frictions on state-land leasing are administrative rather than statutory: revenues on ceded-land parcels carry an OHA share; DHHL dispositions answer to beneficiary priorities; ADC's mission is agricultural; and no agency is tasked with a renewable-siting program. Two current processes bear on this: the legislature commissioned a state-lands renewable study in 2025 (HB 778), and HECO is holding renewable-energy-zone workshops on government lands in July 2026.

10 How the rules got here

A short chronology of the operative provisions, for reference; the repository notes contain the full legislative record, including testimony and committee reports for each act.

Table 8. Sources of the current rules

Year	Instrument	Provision
1976	Act 199 (HRS §205-4.5)	Enumerated-use regime on A/B soils; uses not lis 6 special permit
1980	Act 24	Wind added as a permitted use on agricultural lar no cap; language substantively unchanged since
2008	Act 31	Solar added as a permitted use on D/E soils
2011	Act 217 (SB 631)	Solar extended to B/C soils with the min(10% of p class A excluded. The 10% term originated in Ser 20-acre term in House committee (sized to 5 MW for utility competitive-bid waivers)
2014	Acts 52, 55	SUP path above the cap on B/C soils, with the ag decommissioning security, and restoration condit
2015–2025	Seven amendments	Other uses added to the district (hydro, wireless, camps, landfills); solar provisions unchanged sinc
2021	Honolulu assessment practice; Bill 39	Industrial reclassification of solar-farm land; 80% exemption
2025	Honolulu Ordinance 25-2	Current wind setback, waiver, and repowering sta

The testimony record for the solar acts shows the caps were requested by agricultural and planning agencies and the Farm Bureau, with landowners and the solar industry testifying for wider permissions; the wind setback ordinance responded to sustained community opposition at Kahuku. That history is documented in the repository (`notes/sb631-2011.md`, `notes/acts-2014-2022.md`, `notes/wind-setbacks.md`, and related files) and is not developed further here.

11 Data and methods

GIS: Land Study Bureau soil polygons, State Land Use Districts, and county parcels from the state geoportal; ownership from the Honolulu RPAD OWNALL bulk table (99% acreage coverage, entity-resolved); transmission lines from HIFLD and OpenStreetMap, cross-validated (the 46 kV network is under-mapped, so distances are upper bounds); slope from the USGS 3DEP 1/3-arc-second DEM, computed at 10 m in EPSG:26904 with all layers rasterized to a common grid (raster/vector totals reconcile within one acre). Soil-class acreage totals match the Office of Planning's published LSB table to +0.2%. Legal sources: HRS current text and archived versions (statute dating by snapshot diff), Session Laws of Hawaiʻi scans, Honolulu ordinances and council records, complete LUC SP-docket enumeration from files.hawaii.gov. Transmission unit costs: MISO Transmission Cost Estimation Guide (MTEP24/25), with an inferred Hawaiʻi construction multiplier flagged UNVERIFIED. Screens not applied: interconnection capacity (confidential), flood, aspect, cultural resources, and county-permit-level constraints. Parcel-level tables (`data/oahu_owner_class_transmission.csv`, `data/oahu_parcel_slope.csv`, `data/sup_census.csv`) are TMK-keyed and analysis-ready.

Known gaps: 46 kV line completeness; PVWatts/SAM aspect runs at 21°N; whether co-farmed solar land retains agricultural dedication for tax purposes; Maui County permit archives (manual pull); the pending SP26-417 decision (~Aug 2026).

Background paper assembled 2026-07-12 by Claude (research agent) for M. Roberts. Every factual claim traces to a primary source cached in the repository; claims that do not are tagged UNVERIFIED in the notes files and flagged or excluded here. Repository notes: sb631-2011 · wind-statutory-entry · wind-setbacks · pre2011-solar-bills · acts-2014-2022 · sup-census · property-tax-wedge · cap-quantification · oahu-transmission-screen · oahu-ownership · oahu-slope-screen · slope-cost-literature · state-land-solar · synthesis-2026-07-11.